01-03-2026

Agenda:

Data visualization

python
numpy
pandas

Analysis → 2 years of data
→ 10 days/month
→ 12 x 2 x 10
→ 240

print(240 data points) → plot (visualization) → 240 point

→ To remember 20 people phone.no (1)
→ To remember 20 people faces (2)

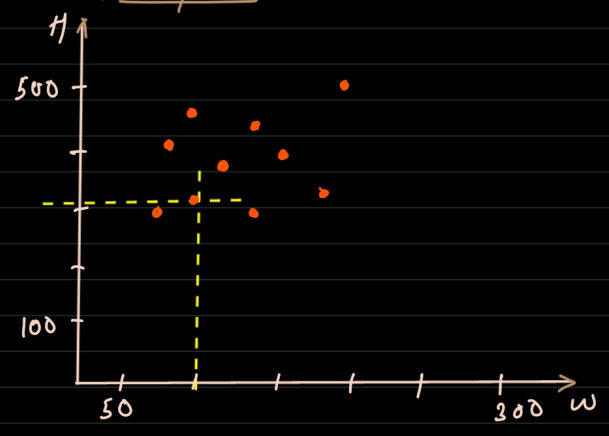
visual elements: — charts, graphs, map, tools
↳ we → see patterns, trends, outliers in large dataset.

Task-1 → problem statement (Business)
↓
Technical knowledge
↓
summary stats, visualization

principles of good visualization

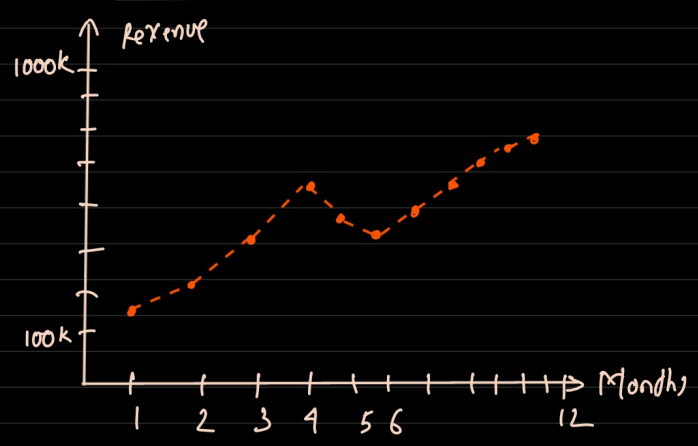
* Clarity → easy to interpret
* proper scale
* Minimalism
* label Everything
* Color wisely
* choose the correct chart for your data.

Scatter plot :—



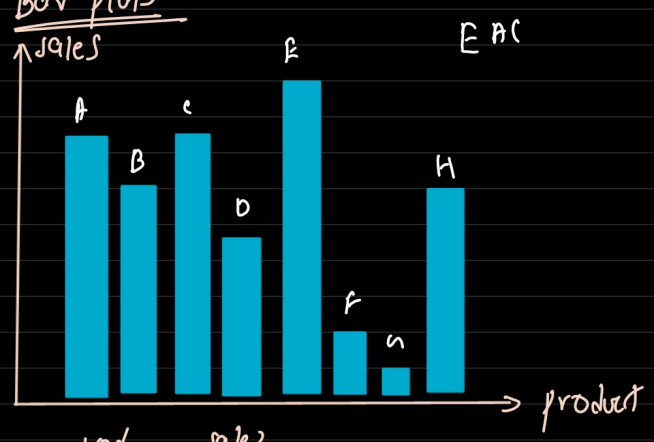
P.ID	w	H
0	100	300
1	120	350
2	98	400

line chart



Month	Income_revenue
Jan	120000$
Feb	140 k $
Mar	180 k $

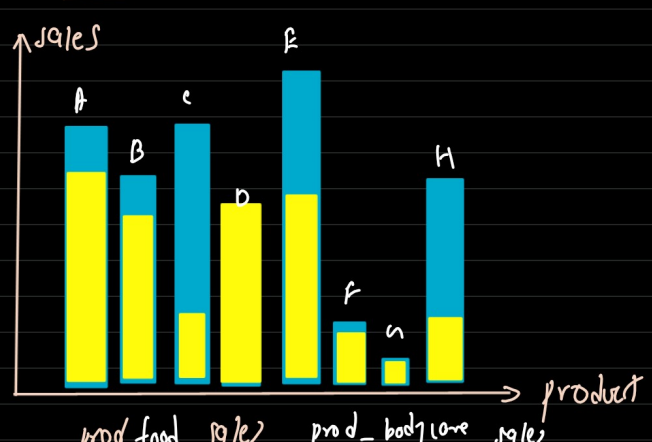
Bar plots



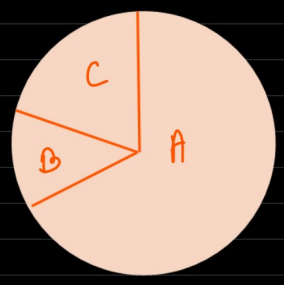
E AC

prod	sales
A	$
B	$
C	$

Stacked Bar plot



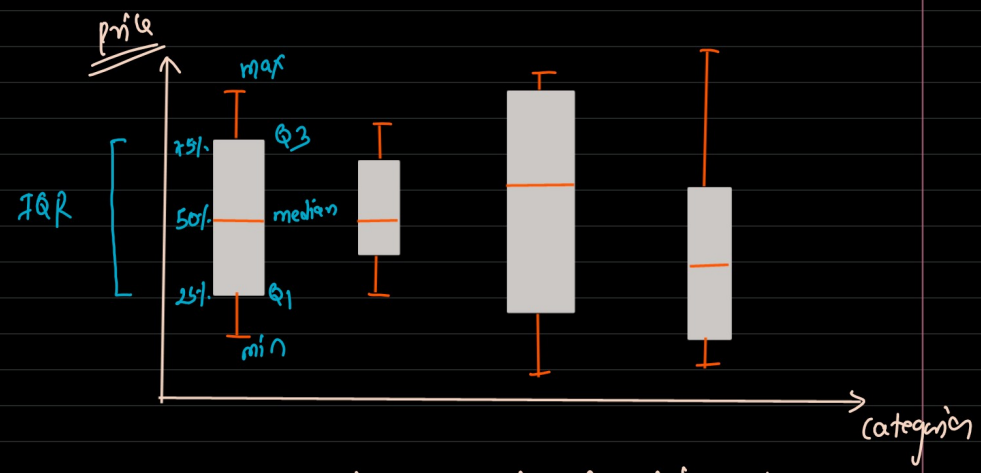
prod_food	sales	prod_bodylove	sales
A	$	A	$
B	$	B	$
C	$	C	$



pie chart

A category → 70
B category → 10
C category → 20

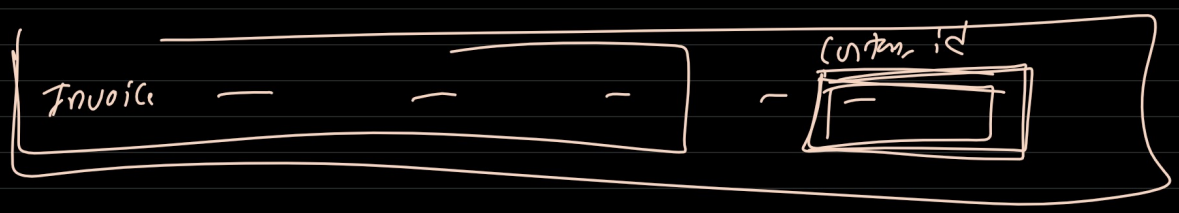
Box plot / whisker plot



price_C1 price_C2 price_C3 price_C4

→ max
→ median
→ min
→ outliers
→ distribution

food → icecream
→ frozen dere
→ frozen _veg
→ gold leaf icecream → 20



get all rows without C

123
456
C123
456
C489

C123
C489

opposite of selecting invoice start with C

invoice that does not start with C

Name	Country	Salary
A	IND	1000
B	US	2000
C	CA	2000
D	Africa	3000
E	US	1000
F	France	4000

df → 6 rows

groupby

country → IND, US, CA, Africa, France

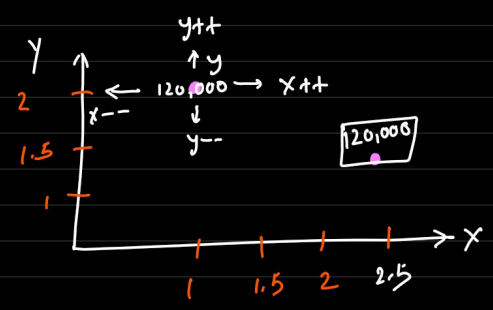
A	IND	1000	1000
B	US	2000	3000
E	US	1000	
C	CA	2000	2000
D	Africa	3000	3000
F	France	4000	4000

Q. find monthly revenue

x, y → (1,2)

x, y

x, y



Text → x, y

Revenue → x, y

120,000

x, y

↓ ↓

2.5 1.5

x → 2.5

y → 1.8 > 1.5

y → 3.5

Text